ERP & Industry 4.0
Assignment

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Section: 3

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Industry Overview

The e-commerce industry has transformed the way consumers purchase products by introducing multiple channels for online shopping such as websites and mobile applications. Companies such as Amazon, Flipkart, Myntra and Meesho manage millions of transactions every day. At the same time, they coordinate with customers, suppliers, warehouses, logistics partners and financial institutions.

Managing such complex operations manually is difficult, inefficient and leads to errors. Enterprise Resource Planning (ERP) systems connect all business functions on a single platform. This allows smooth data flow across departments. A good ERP system helps an e-commerce business in improving customer satisfaction, optimize inventory and procurement, manage finances and support decision-making.

1. Required ERP Modules

The following ERP modules are necessary for an e-commerce company. The five most critical modules are ranked below based on their business importance.

Module	Purpose	Justification
1. Website	Allows customers to browse products, place orders, make payments and track deliveries	Acts as a primary user interface which enables smooth and seamless online shopping experience for the customers
2. Sales and order management	Manages quotations, sales orders, deliveries and invoices	It ensures accurate order processing and timely fulfilment by automating the order-to-cash process
3. Inventory	Manages warehouses, stock movements, receipts and deliveries	Provides real-time stock visibility, automates inventory updates during receipts and deliveries and maintains optimal stock levels
4. Procurement	Handles vendor management and purchase orders along with receipts	Streamlines the process by automating purchase orders and ensuring timely replenishment based on demand
5. Finance & Accounting	Handles billing, customer invoices, payments, taxes, accounts payable and accounts receivable	Supports real-time financial visibility, enables accurate financial transactions and ensures regulatory compliance

Other supporting ERP Modules:

- Employees
- Dashboards
- Contacts
- CRM

2. Master Data

Master data have to be configured before any business transactions can begin.

Master Data	Description	Importance
Customer	Stores all customer details like contact information, delivery address, billing address and order history	Enables order processing and customer communication
Product	Includes product code, price, product description, taxes, barcode	Forms the basis for sales, inventory and reporting
Supplier	Contains supplier information, lead times and payment terms	Supports procurement activities
Warehouse	Defines warehouses, stock locations, packing stations and shipping locations	Enables warehouse operations and inventory tracking
Product Categories	Forms groups for pricing, taxation, reporting and inventory valuation	Simplifies reporting and pricing
Payment methods	Configures online payment pathways like Razorpay, Paypal, UPI and Cash on Delivery	Supports online payment processing
Delivery methods	Stores courier partners, shipping charges, delivery rules and tracking integration	Facilitates order fulfillment
Tax configuration	Defines GST rates and tax rules applicable to different product categories	Ensures compliance
Employee	Stores employee details for warehouse staff, procurement officers etc and department roles	Supports HR and workflow approvals

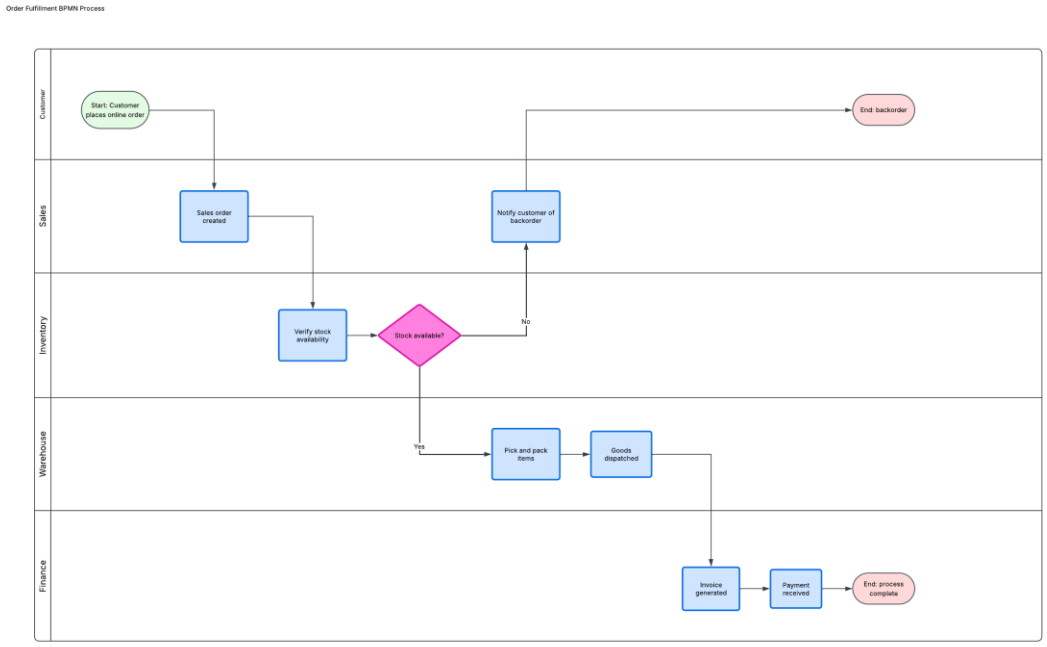
3. Business Transactions

a) Customer Order Processing (Order-to-Cash)

Departments involved:

- Customer
- Sales
- Warehouse
- Finance
- Inventory

Description: This transaction begins when a customer places an order through the e-commerce website. The Saler department creates the sales order, Inventory verifies product availability, Warehouse picks, packs and dispatches the goods, and Finance generates the invoice and records the payment. If the product is out of stock, the customer is notified about the backorder.



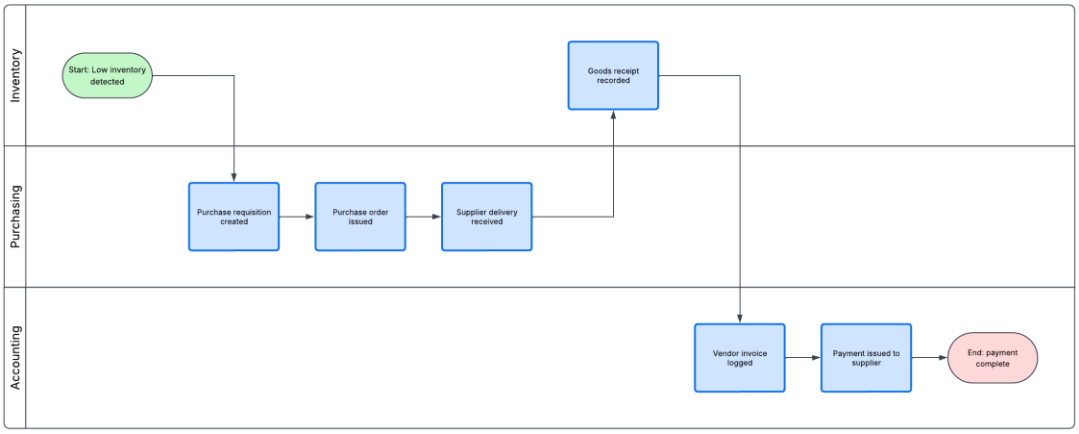
b) Procurement (Procure-to-Pay)

Departments involved:

- Inventory
- Purchasing
- Accounting

Description: When inventory falls below the reorder level, the Inventory department triggers a purchase requisition. The Purchasing department issues a purchase order to the supplier. After the supplier delivers the goods, Inventory records the receipt. The Accounting dept then verifies the vendor invoice and processes payment to the supplier.

Procure-to-Pay BPMN Diagram



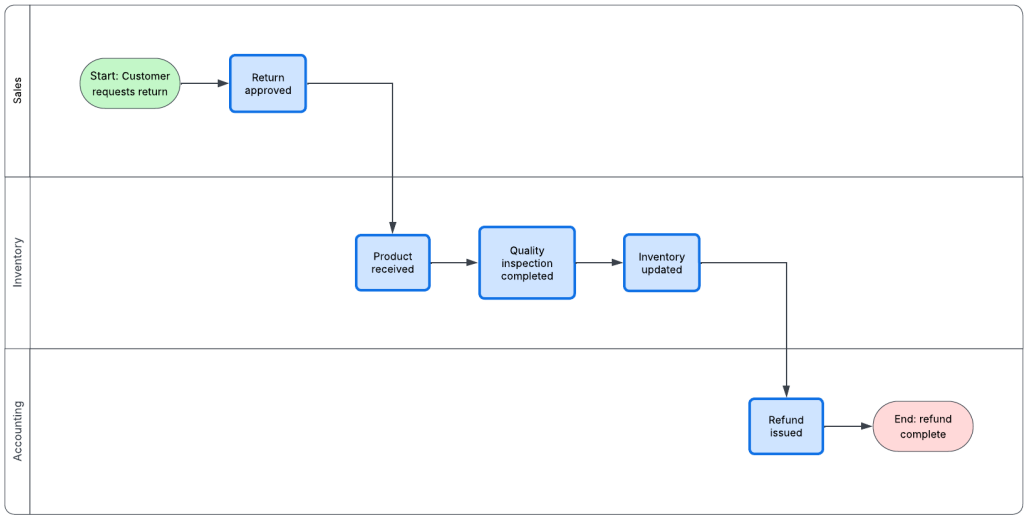
c) Customer Return Process

Departments involved:

- Sales
- Inventory
- Accounting

Description: When a customer requests a product return, the Sales department reviews and approves the request. The returned product is received by the Inventory department and undergoes quality inspection. After inventory records are updated, the Accounting department issues the customer refund, completing the return process.

Sales Return Process BPMN

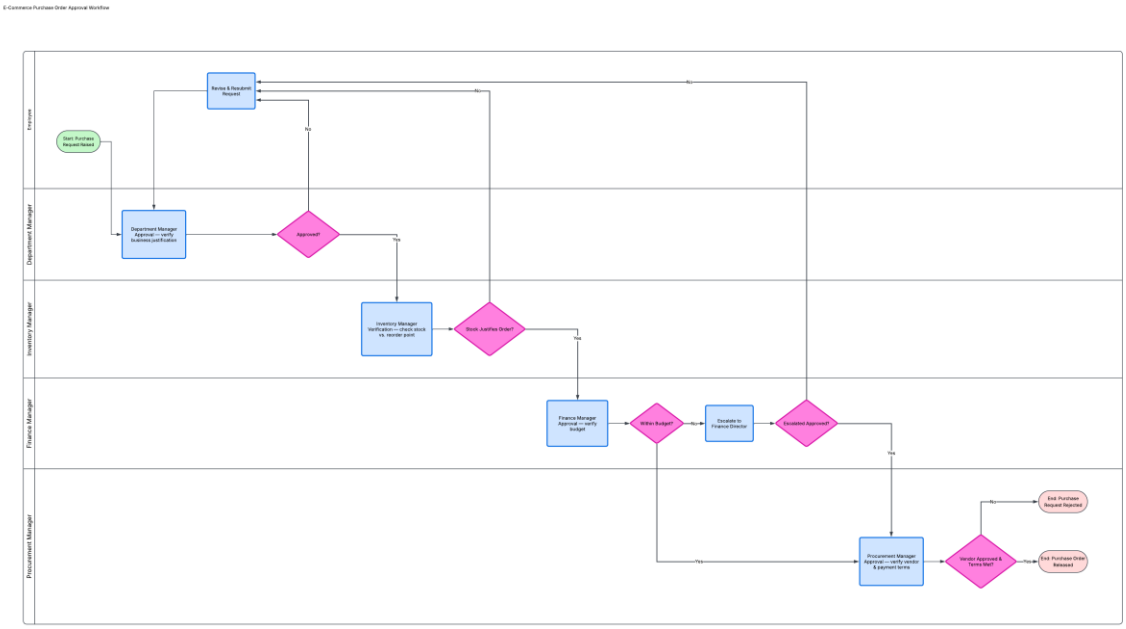


4. Critical Workflow

Purchase order approval workflow

In an e-commerce company, stock must be replenished continuously to avoid stockouts, yet uncontrolled purchasing impacts the margin negatively. A structured purchase order (PO) approval workflow ensures spend is verified against actual stock levels, budget, vendor terms before any transaction takes place.

Approval Stage	Decision Criteria
Department manager	Confirms business requirement
Procurement manager	Approves supplier selection, purchasing terms and pricing
Finance manager	Confirms budget availability
Inventory manager	Verifies stock availability before purchasing



Approval rules by value threshold (example)

PO Value (INR)	Required approvals
<100000	Department manager only
100000-1000000	Department manager → Inventory manager → Finance manager
1000000-5000000	Full chain
>5000000	Full chain + CFO sign-off

Additional Rules

- Any rejection at any step sends the PR back to the originating employee for revision and resubmission.

- Each approver has 24 to 48 hours to act before the request auto-escalates to their manager to prevent any bottlenecks.
- The person who raises the PR request cannot approve it and Finance manager approval is mandatory for any PO regardless of value to prevent unauthorized transactions.
- Every approval is timestamped and logged against the PO for compliance.

5. Integration points

a) Sales and Inventory

Data exchanged:

- Product information
- Available inventory
- Reserved quantity
- Delivery status
- Stock updates

Importance of Integration: When a customer places an order, the Inventory module must immediately reserve the required stock so that the same unit is not sold to two customers at once. Once the order is shipped, inventory quantities should be automatically updated so that the website always shows the true current stock count.

Consequences if this integration fails:

- Lost sales
- Delayed deliveries
- Customer dissatisfaction
- Overselling products

b) Procurement and Finance

Data exchanged:

- Purchase orders
- Tax records
- Payment information
- Vendor information
- Goods receipt notes

Importance of Integration: Once goods are received from the supplier, Accounts payable needs the Goods receipt note and vendor invoice to flow through automatically so that the payment is issued on time and matched correctly to what was actually delivered. This ensures accurate financial reporting.

Consequences if this integration fails:

- Duplicate supplier payments
- Delayed vendor payments
- Incorrect financial statements

Conclusion

An integrated ERP system assists an e-commerce company to streamline operations by connecting different departments like sales, procurement, finance, logistics and customer service on a single platform. The proposed ERP design enables real-time inventory visibility, accurate financial transactions, efficient and smooth order processing and also helps in better decision-making through integrated data. This conceptual ERP design can be effectively implemented using an ERP platform such as Odoo.

By standardizing workflows and connecting all departments, a company can improve its operational efficiency, enhance customer satisfaction, minimize costs and also support sustainable business growth.